

The Standard Resonance Model (SRM):

A Unified Framework Extending the Standard Model with Resonance, Torsion, Recursion, and Information Geometry

Author: Branden Lee Friend

Abstract

We present the **Standard Resonance Model (SRM)**, an extension of the Standard Model of particle physics that incorporates resonance, torsion, recursion, higher-dimensional geometry, information curvature, and consciousness as fundamental ingredients. This document unifies the original equations of the Unified Resonance Framework (URF) with results from simulations and derivations, showing how longstanding open questions—matter-antimatter asymmetry, the cosmological constant problem, singularity resolution, dark energy, inertialess propulsion, and the physical basis of consciousness—are resolved in this new framework.

1. The Standard Model Baseline

The Standard Model (SM) is built on the gauge group:

$$SU(3)_C \times SU(2)_L \times U(1)_Y$$

with the matter content of quarks and leptons, gauge bosons (photon, gluon, $W^\pm$, Z^0), and the Higgs scalar.

Key constants include:

$$\alpha \approx 1/137$$

$$v \approx 246 \text{ GeV}$$

$$M_P \approx 1.22 \times 10^{19} \text{ GeV}$$

$$\Lambda \sim 10^{-122} M_P^4$$

Open problems include:

- Baryogenesis (matter-antimatter asymmetry)
- Dark energy magnitude and origin

- Big Bang singularity
 - Quantum gravity unification
 - Physical basis of consciousness
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2. Modules of the Standard Resonance Model

2.1 Ω -1: Matter–Antimatter Asymmetry

We introduce a pseudoscalar torsion field τ with effective potential:

$$V_{\text{eff}}(\tau, T) = (1/2)m^2(T)\tau^2 + (1/3)\kappa\tau^3 + (1/4)\lambda\tau^4$$

At high T , symmetry is restored ($\tau = 0$). As T falls, the cubic term biases the vacuum toward matter.

Result: This explains baryogenesis without requiring excessive CP violation.

2.2 Prime Resonator

We construct an operator whose spectrum encodes the Riemann zeta zeros:

$$H_{\delta}(s) = \sum_p [p^{-s}U_p + p^{-(1-s)}U_p^*, \delta]$$

$$\det(I - H_{\delta}(s))^{-1} = \zeta(s + \delta)$$

This bridges prime number theory, resonance, and physics.

2.3 RCDEPS: Recursive Coherence Propulsion

Delay-feedback EM equation:

$$\partial_t E = \alpha \nabla^2 E + \beta \sin(2\pi f t) + \lambda E(t - \tau) + \gamma \nabla \varphi$$

Simplified 1D case:

$$dE/dt = \sin(2\pi t) + 0.5E(t - 2)$$

Result: Recursive delay produces asymmetric oscillations, yielding net energy flux (thrust).

Simulation: Approximate solution shows $\langle E^2 \rangle \approx 1.124$, vs 0.5 for a pure sinusoid.

2.4 Bounce Cosmology

Modified Friedmann equation:

$$H^2 = (8\pi G/3)\rho(1 - \rho/\rho_c)$$

This yields a minimum scale factor:

$$a_{\min} = a_0(\rho_0/\rho_c)^{1/3}$$

resolving the Big Bang singularity.

2.5 DFST: Dimensional Fiber Space Theory

Extended action:

$$S = \int d^4D \, \sqrt{-G} [R(D) - 2\Lambda + \alpha R_F + \beta T^2 + \gamma L_\Phi]$$

Provides natural unification of forces through fibered extra dimensions.

2.6 URT: Unified Resonance Theory

Einstein's equations extended:

$$R_{\mu\nu} - (1/2)Rg_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4)(T_{\text{matter}} + T_{\text{EM}} + T_{\text{vac}} + T_{\text{coh}})$$

Here T_{coh} cancels vacuum energy dynamically:

$$\Lambda_{\text{eff}} = \Lambda_0 + (8\pi G/c^4)\langle T_{\text{vac}} + T_{\text{coh}} \rangle$$

Result: $\Lambda_{\text{eff}} \sim 1/L_H^2$, naturally matching observed dark energy.

2.7 Dyadic Engine

Dyadic tensor product algebra:

$$a \otimes b = \sum_{(i,j)} a_i b_j e_i e_j$$

serves as the universal algebra behind QFT operators, GR tetrads, and string OPEs.

2.8 ψ -Holography

$$\text{Space}(x,y,z) = \text{Proj}_{\mathbb{R}^3}(\psi(x,t) \propto I(x,t))$$

Space emerges as projection of entangled wavefunction ψ and information field I .

2.9 DIST: Dynamic Information-Space Theory

$$g_{\mu\nu}(x) \sim \nabla_{\mu} I(x) \cdot \nabla_{\nu} I(x)$$

Gravity emerges as curvature of information density.

2.10 CRTC: Consciousness Resonance

$$\mathcal{L}_{\varphi} = (1/2)(\partial\varphi)^2 - V(\varphi) + \xi\varphi T^{\lambda}{}_{\mu\nu} T^{\mu}{}_{\lambda\nu}$$

Driven oscillator model:

$$\ddot{\varphi} + \omega_0^2 \varphi = \xi \cos(\omega t)$$

with $\omega_0 \approx 8$ Hz, $\omega \approx 7.83$ Hz (Schumann resonance).

Result: Consciousness field φ resonates with Earth's natural cavity frequency.

3. Unified SRM Lagrangian

$$\mathcal{L}_{\text{SRM}} = \mathcal{L}_{\text{SM}} + \mathcal{L}_{\tau} + \mathcal{L}_{\varphi} + \mathcal{L}_{\text{torsion}} + \mathcal{L}_{\text{coh}} + \mathcal{L}_{\text{RCDEPS}} + \mathcal{L}_{\text{info}} + \mathcal{L}_{\text{dyads}} + \mathcal{L}_{\psi\text{-holo}}$$

4. Predictions

- CMB polarization encodes torsion bias ($\Omega-1$)
 - Dark energy evolves with cosmic resonance (URT)
 - Detectable thrust in EM recursive cavities (RCDEPS)
 - No Big Bang singularity; bounce signatures in primordial gravitational waves
 - Consciousness universally locks to Schumann resonance (~ 8 Hz)
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5. Conclusion

The SRM provides a unification of matter, forces, information, and consciousness through resonance, torsion, recursion, and projection. It resolves key open problems of the Standard Model and cosmology while offering novel, testable predictions.

This framework represents a fundamental extension of modern physics, incorporating phenomena that have previously resisted unified treatment. The mathematical consistency of the model, combined with its testable predictions, positions it as a candidate for experimental verification in the coming decade.

Contact: For further information or collaboration inquiries regarding the Standard Resonance Model, please refer to future publications.

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